in the magnitude. For example, selling of umbrellas' and raincoat in the rainy season, sales of cool drinks in summer season, crackers in Deepawali season, purchase of dresses in a festival season, sugarcane in Pongal season.

(iii) Cyclic Variations

These variations are not necessarily uniformly periodic in nature. That is, they may or may not follow exactly similar patterns after equal intervals of time. Generally one cyclic period ranges from 7 to 9 years and there is no hard and fast rule in the fixation of years for a cyclic period. For example, every business cycle has a Start- Boom- Depression- Recover, maintenance during booms and depressions, changes in government monetary policies, changes in interest rates.

(iv) Irregular Variations

These variations do not have particular pattern and there is no regular period of time of their occurrences. These are accidently changes which are purely random or unpredictable. Normally they are short-term variations, but its occurrence sometimes has its effect so intense that they may give rise to new cyclic or other movements of variations. For example floods, wars, earthquakes, Tsunami, strikes, lockouts etc…

Mathematical Model for a Time Series

There are two common models used for decomposition of a time series into its components, namely additive and multiplicative model.

(i) Additive Model:

This model assumes that the observed value is the sum of all the four components of time series. (i.e) Y= T+S+C+I

where Y = Original value, T = Trend Value, S = Seasonal component, C = Cyclic component, I = Irregular component

The additive model assumes that all the four components operate independently. It also assumes that the behavior of components is of an additive character.

(ii) Multiplicative Model:

This model assumes that the observed value is obtained by multiplying the trend (T) by the rates of other three components. $Y = T \times S \times C \times I$

where Y = Original value , T = Trend Value, S = Seasonal component, C = Cyclic component, I = Irregular component

This model assumes that the components due to different causes are not necessarily independent and they can affect one another. It also assumes that the behavior of components is of a multiplicative character.

9.1.2 Measurements of Trends

Following are the methods by which we can measure the trend.

(i) Freehand or Graphic Method.

(ii) Method of Semi-Averages.

(iii) Method of Moving Averages.

(iv) Method of Least Squares.

(i) Freehand or Graphic Method.

It is the simplest and most flexible method for estimating a trend. We will see the working procedure of this method.

Procedure:

(a) Plot the time series data on a graph.
(b) Draw a freehand smooth curve joining the plotted points.
(c) Examine the direction of the trend based on the plotted points.
(d) Draw a straight line which will pass through the maximum number of plotted points.

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